sEKV Parameter Extraction for the Generic 180nm Process

Check extraction result on nMOS and pMOS of various sizes (Version 1)

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In this notebook we check the sEKV parameters for nMOS and pMOS transistors of different sizes. We use data generated with the EKV 2.6 compact model for a generic 180nm CMOS process.

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1 Introduction

In this notebook we will compare the sEKV model and the simulations with the EKV 2.6 compact model for 3 nMOS and pMOS transistors having all the same width $W = 10\ \mu m$ and different lengths $L = 10\ \mu m$, $1\ \mu m$ and $180\ nm$. For the sEKV model we will use the parameters that have been extracted for nMOS transistors for a generic 180nm bulk CMOS process. The physical parameters are given in Table 1.1, the global process parameters in Table 1.2 and the transistor parameters are given in Table 1.3.

Table 1.1: Physical parameters

Parameter	Value	Unit
T	300	K
U_T	25.875	mV

Table 1.2: Global process parameters

Parameter	Value	Unit
V_{DD}	1.8	V
C_{ox}	8.46	$\frac{fF}{\mu m^2}$
W_{min}	200	nm
L_{min}	180	nm

Table 1.3: Transistor process parameters

Parameter	NMOS	PMOS	Unit
sEKV parameters			
n	1.27	1.31	-
$I_{spec\square}$	715	173	nA
V_{T0}	0.455	0.457	V
L_{sat}	39	12	nm
λ	20	20	$\frac{V}{\mu m}$
Overlap capacitances parameters			
C_{GDo}	0.366	0.328	$\frac{fF}{\mu m}$
C_{GSo}	0.366	0.328	$\frac{fF}{\mu m}$
C_{GBo}	0	0	$\frac{fF}{\mu m}$
Junction capacitances parameters			
C_J	1	1.121	$\frac{fF}{\mu m^2}$
C_{JSW}	0.204	0.248	$\frac{fF}{\mu m}$
Flicker noise parameters			
K_F	8.1e-24	6.8e-23	J
AF	1	1	-
ρ	0.05782	0.4818	$\frac{V \cdot m^2}{A \cdot s}$
Matching parameters			

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Table 1.3: Transistor process parameters

Parameter	NMOS	PMOS	Unit
A_{VT}	5	5	$mV \cdot \mu m$
A_β	1	1	$\% \cdot \mu m$
Source and drain sheet resistance parameter			
R_{sh}	600	2386	$\frac{\Omega}{\mu m}$
Width and length parameters			
ΔW	39	54	nm
ΔL	-76	-72	nm

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2 nMOS

2.1 IC versus $V_G - V_{T0}$

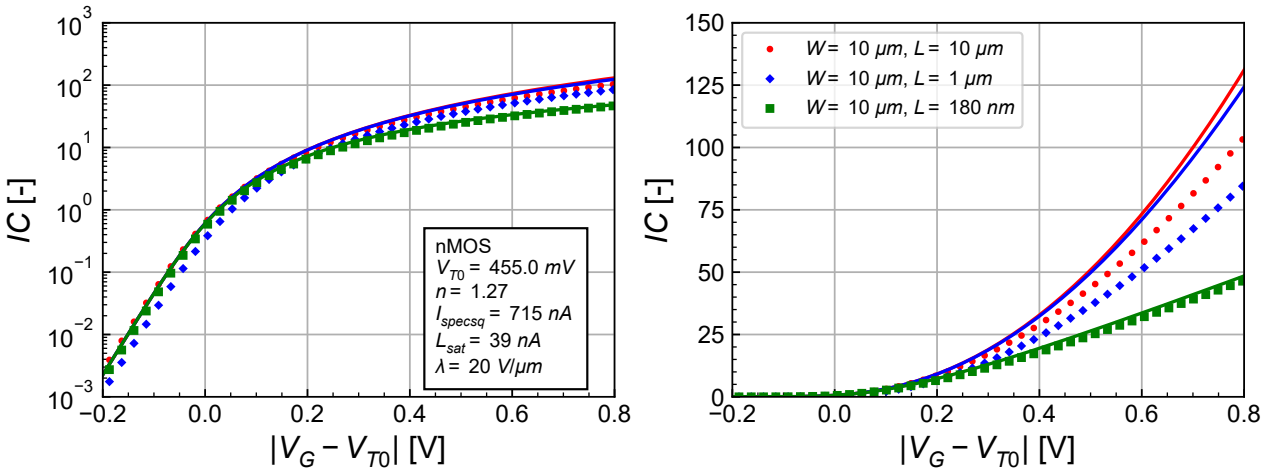


Figure 2.1: IC versus $V_G - V_{T0}$ of the sEKV model compared to simulations with EKV 2.6 for three different lengths.

The fit is good in weak and moderate inversion for the short and long transistor, however the medium is off (threshold voltage too large). The fit in strong inversion is good for the minimum length transistor but is a bit off for the long and medium channel transistors. This is certainly due to the mobility reduction due to the vertical field which is not accounted for in sEKV.

2.2 G_m and G_m/I_D versus IC

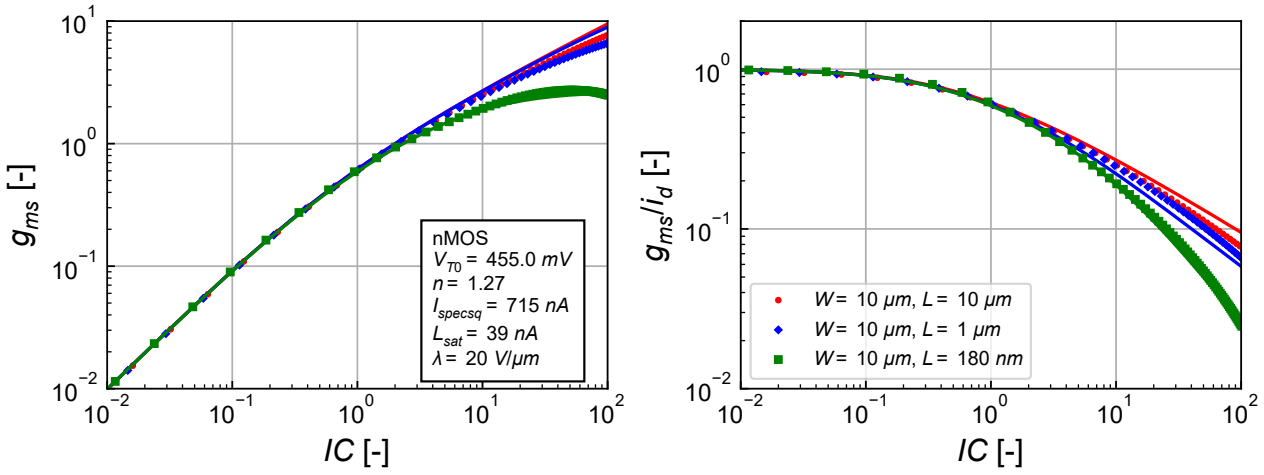


Figure 2.2: g_{ms} and g_{ms}/i_d versus IC of the sEKV model compared to simulations with EKV 2.6 for three different lengths.

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The fit is good in weak and moderate inversion for all lengths. The fit is good in strong inversion for the minimum length transistor, buta bit off for the long and medium length transistors. This is again due to the mobility reduction due to the vertical field which has an impact on g_{ms} and g_{ms}/IC.

2.3 G_{ds} versus V_D

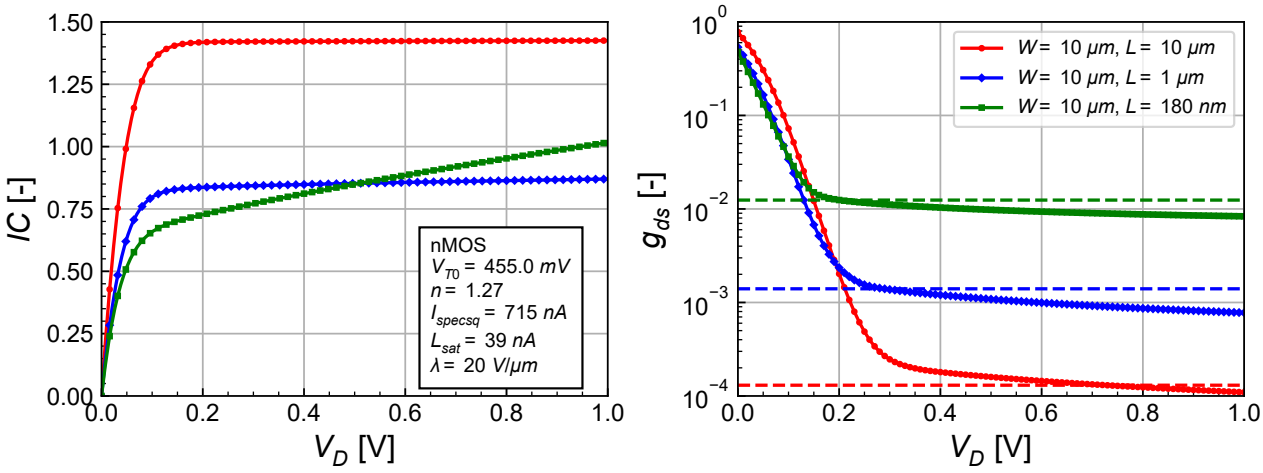


Figure 2.3: Approximation of the output conductance in saturation for three different lengths.

The approximation of the output conductance is OK for the long-channel transistor and a bit overestimated for the medium and short channel transistors.

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3 pMOS

3.1 IC versus $V_G - V_{T0}$

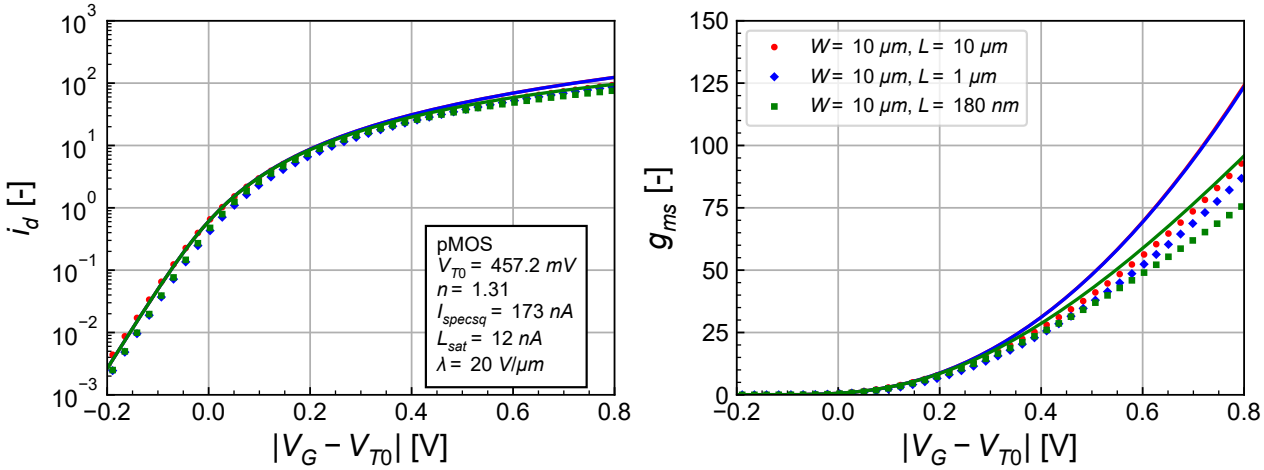


Figure 3.1: IC versus $V_G - V_{T0}$ of the sEKV model compared to simulations with EKV 2.6 for three different lengths.

The fit is good in weak inversion for the long-channel transistor, however the threshold voltage is too large for the medium and short-channel transistors. This is due to charge-sharing and reverse short-channel effects which increase the threshold voltage for small transistors.

3.2 G_m and G_m/I_D versus IC

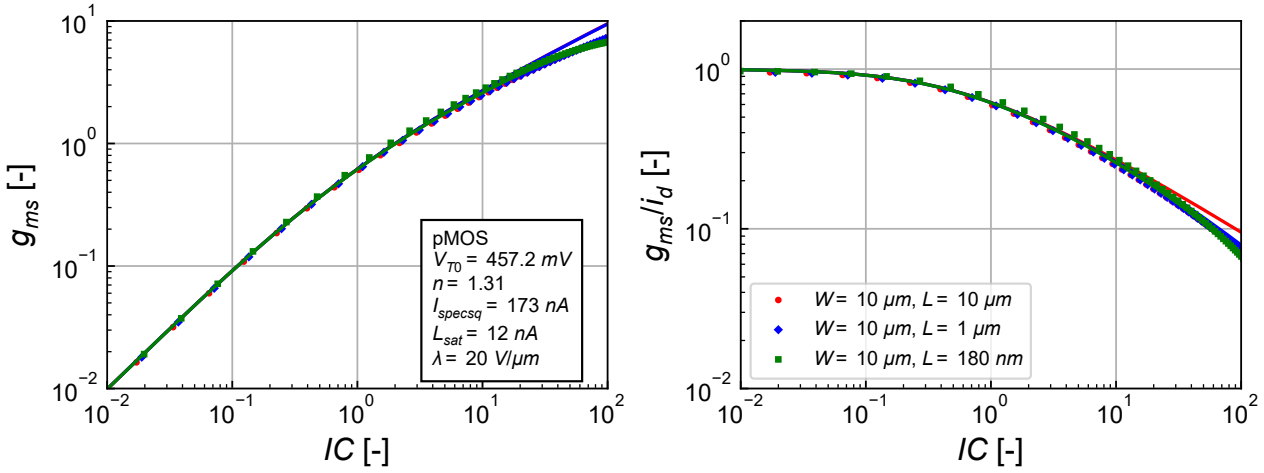


Figure 3.2: g_{ms} and g_{ms}/i_d versus IC of the sEKV model compared to simulations with EKV 2.6 for three different lengths.

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The fit of the normalized source transconductance g_{ms} is good in weak and moderate inversion for all lengths. g_{ms} is a bit overestimated in strong inversion for the long and medium channel transistors. The fit of the normalized g_{ms}/IC characteristic is good in weak and moderate inversion for all three lengths. It is a bit overestimated for the long-channel transistor in strong inversion.

3.3 G_{ds} versus V_D

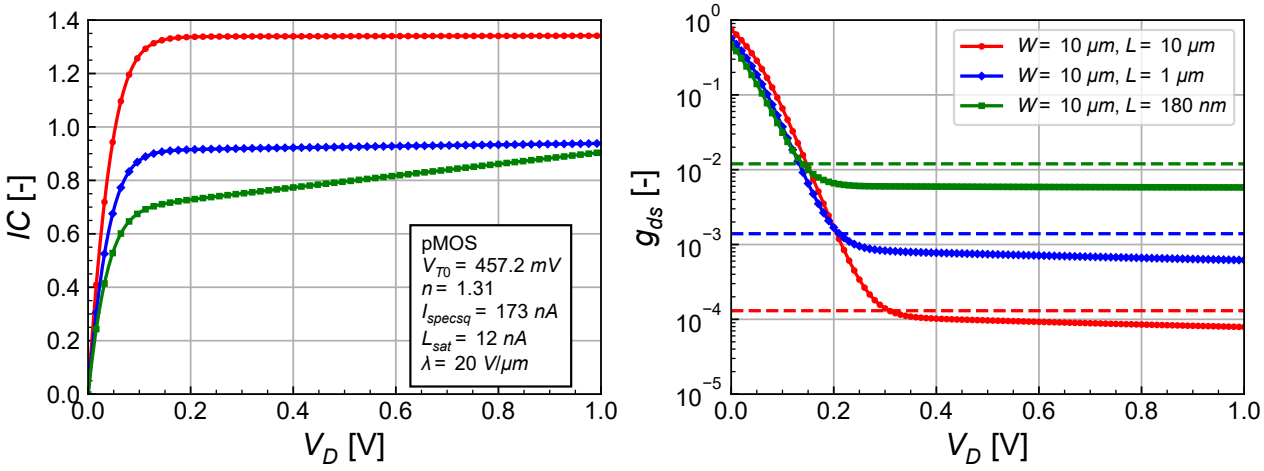


Figure 3.3: Approximation of the output conductance in saturation for three different lengths.

The output conductance is overestimated for all three channel lengths, but particularyl for the short-channel transistors. This will result in an underestimation of the intrinsic gain.

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4 Conclusion

The sEKV parameters have been extracted for n- and p-channel transistors for a generic 180 nm bulk CMOS process. The fit of the IC versus $V_G - V_{T0}$ is acceptable for all lengths and for both nMOS and pMOS. The fit of the normalized source transconductance g_{ms} and g_{ms}/IC characteristic are good for all lengths and for both nMOS and pMOS. The approximation of the output conductance is OK for all lengths for nMOS and a bit overestimated for the pMOS.

We will use these parameters for all the examples in the book.

> **!** Important
>
> The overall fit may seem a bit disapointing. However, we have to remember that the goal of the sEKV model is not to replace the compact model, but rather to have an accurate estimation of the transconductances and of the G_m/I_D characteristic for inversion coefficients IC ranging between 10^{-2} to 10^2.

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